Lecture 4: Chemical Differentiation & Magnetospheres

Planetary Systems

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Aurora australis, ISS. Credit: NASA

Learning objectives

By the end of this lecture, you will be able to:

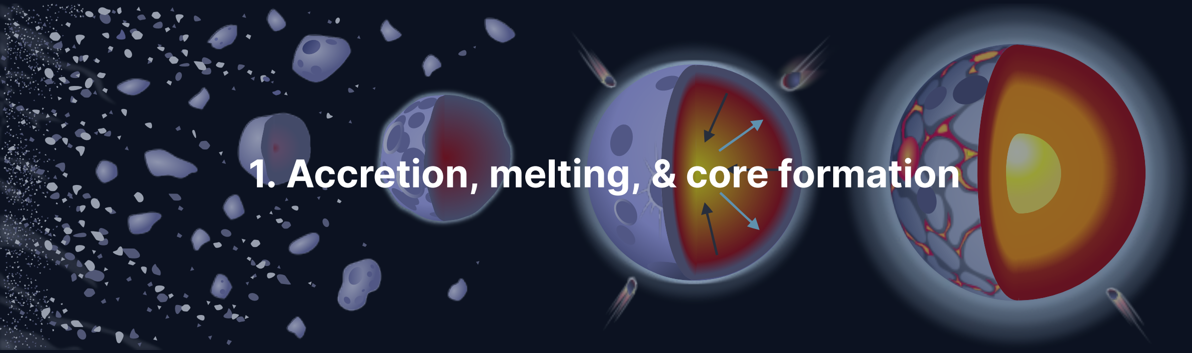
- Explain metal–silicate separation and use siderophile depletion and Hf–W as its evidence
- Derive the magnetic Reynolds number and assess whether a core can run a dynamo
- Compare the magnetic fields and magnetospheres across the solar system

Recap: Lecture 3

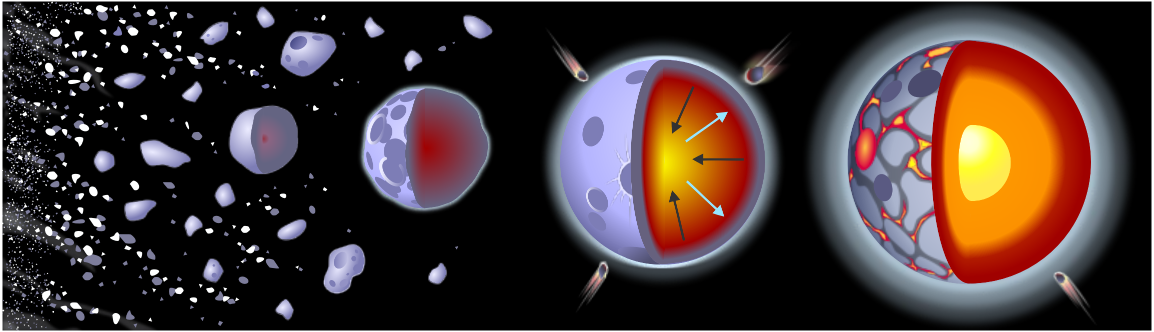
- Four heat sources; conduction too slow for planets, $\tau \sim L^2/\kappa$
- $\mathrm{Ra} \sim 10^7$–10^8 for Earth: vigorous convection, mobile lid against stagnant lid
- Tidal heating: Io, Enceladus, Europa

Today: The *chemical* consequences of melting, and how differentiation enables magnetic fields.

1. Accretion, melting, & core formation

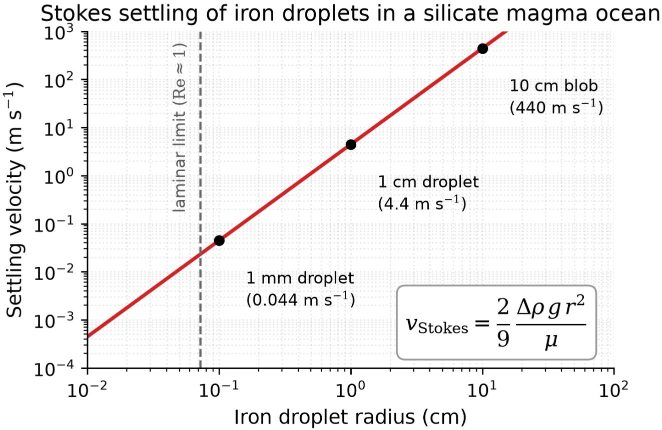


Planetary differentiation. Credit: Wikimedia, CC BY-SA 4.0



Planetary differentiation: a molten body separates into a metal core and a silicate mantle.
Credit: Wikimedia, CC BY-SA 4.0.

Accretional heating and $^{26}\mathrm{Al}$ make a **magma ocean**; iron ($\rho \sim 7000\ \mathrm{kg\,m^{-3}}$) is immiscible with silicate (~3000) and its droplets sink at the Stokes speed $v_{\mathrm{Stokes}} = \frac{2}{9}\,\Delta\rho\, g\, r^2/\mu$.



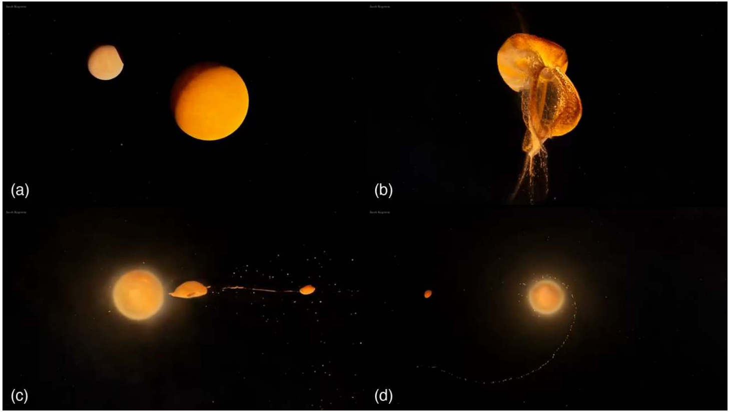
Stokes settling of iron droplets in a magma ocean: $v \propto r^2$, 4.4 m s^{-1} at 1 cm, about 0.5 m s^{-1} with turbulent drag. Course figure.

Iron rain empties a magma ocean of its metal within weeks: on geological clocks, core formation is instantaneous.

The magma ocean state, and why it is essential

- **Liquid mantle** (μ ~ 0.1–1 Pa s, $\mathrm{Ra} \gtrsim 10^{25}$): iron droplets sink at m s^{-1}, a core forms within weeks
- **Solid mantle** (μ ~ 10^{21} Pa s): a km-sized iron body needs longer than the age of the universe to reach the centre
- Magma ocean lifetime 10^3–10^8 yr: long enough for core formation, short for planetary evolution

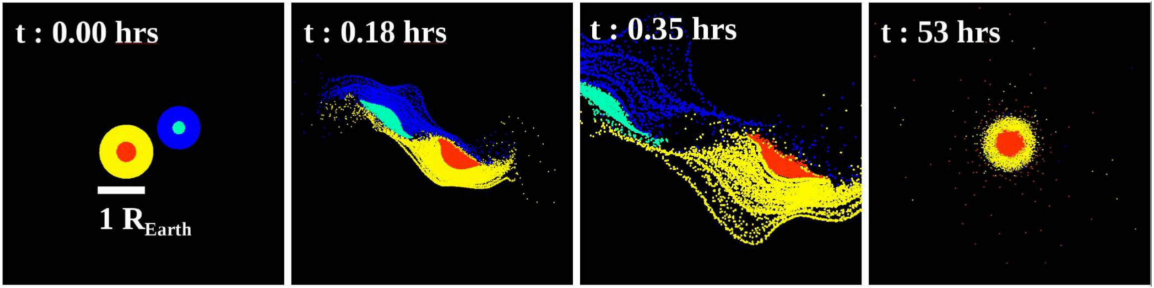
Core formation happens while the planet is molten, or not at all; chondritic asteroids that never melted never formed cores.



SPH simulation of the Moon-forming giant impact: (a) Theia approaches, (b) contact and disruption, (c) fallback, (d) a satellite on a wide orbit. Kegerreis et al. (2022). ▸ Play animation

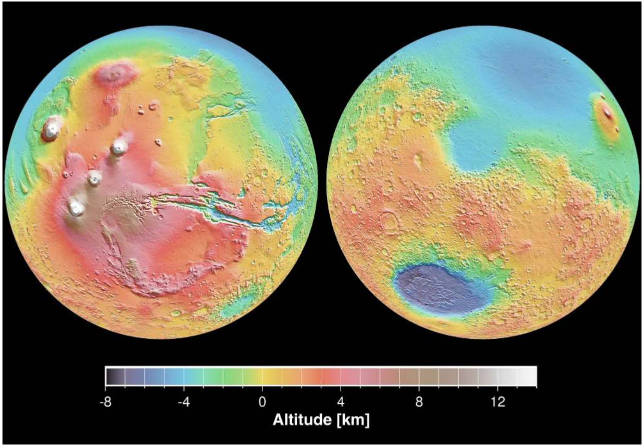
▸ Source video

A Mars-sized Theia hit the proto-Earth about 4.5 Ga, re-melted the mantle into the final deep magma ocean and made the Moon; lunar $\varepsilon^{182}\mathrm{W}$ matches Earth's mantle.



A mantle-stripping giant impact on proto-Mercury: a 2.25 Mercury-mass target hit at b = 0.5 and 30 km s^{-1} keeps 0.95 Mercury masses with Z_{Fe} = 0.67. Chau et al. (2018).

Mercury's core mass fraction of about 74% (Earth 32%, Mars 24%): a hit-and-run impact stripped 50 to 70% of its mantle. Giant impacts set the final core-to-mantle ratio.



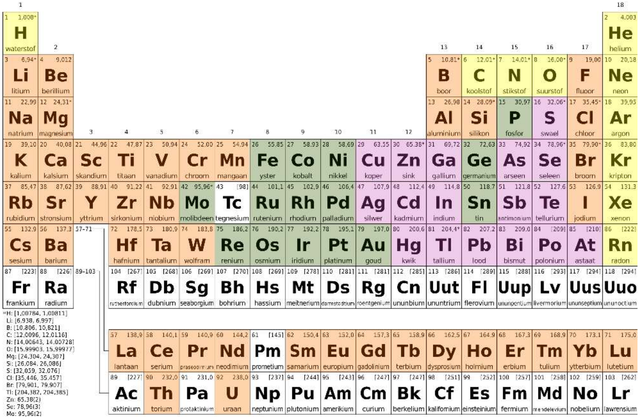
Mars's hemispheric dichotomy in MOLA topography: low northern plains (blue) against cratered southern highlands; Hellas at the right. Credit: NASA/JPL.

The dichotomy (about 6 km in elevation, 30 km in crustal thickness) is the relaxed scar of a Borealis-scale impact: crust moved sideways, not lost.

The late veneer

- Highly siderophile elements (Ir, Pt, Os, Au) are about 200× more abundant in Earth’s mantle than full metal extraction predicts
- Their ratios are **chondritic**, not fractionated
- About 0.5% of Earth’s mass arrived **after** core formation closed; Mars and the Moon show a smaller veneer

Source: Walker (2009); Day et al. (2016)



Periodic table coloured by Goldschmidt class: siderophile (green), lithophile (orange), chalcophile (purple), atmophile (yellow). Credit: Wikimedia Commons, CC BY-SA 3.0.

Siderophile to the metal (Fe, Ni, Pt, Au), lithophile to the silicate (Si, Mg, Al, U, Th), chalcophile to sulfide, atmophile to gas: the siderophile depletion of the mantle is **direct evidence** for core formation.

Partition coefficients, and why silicon is in the core

$$D_i^{\mathrm{met/sil}} \equiv C_i^{\mathrm{metal}} / C_i^{\mathrm{silicate}}$$

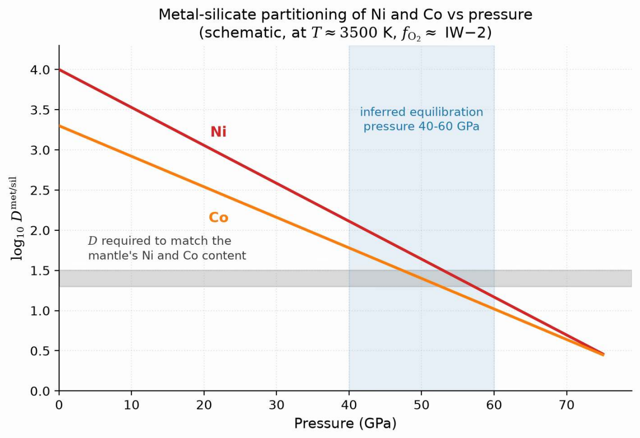
Element	$D^{\mathrm{met/sil}}$	Behaviour
Ir, Os, Pt	10^4–10^6	Highly siderophile
W, Mo	10^2–10^4	Moderately siderophile
Ni, Co	$10^4 \rightarrow 10^2$	Moderately siderophile
Fe	~10	Weakly siderophile
Si, Mg, O	$\ll 1$	Lithophile

Ni, Co: D falls from 10^4 at 1 GPa to 10^2 at 50 GPa.

- D depends on pressure, temperature and **oxygen fugacity** (the effective O_2 partial pressure of the melt)
- Si is lithophile below 10 GPa, but $D_{\mathrm{Si}} \rightarrow 1$ at 40–60 GPa: the core holds several wt% Si, part of its 5–10% density deficit (Lecture 8)

Deep magma ocean partitioning at 40–60 GPa is the standard model for Earth's core composition.

Source: Rubie et al. (2015); Fischer et al. (2015); Hirose et al. (2021); Shahar et al. (2026)



Metal-silicate partition coefficients of Ni and Co against pressure; the grey band is the value the mantle's Ni and Co require. Course figure, after Siebert et al. (2013).

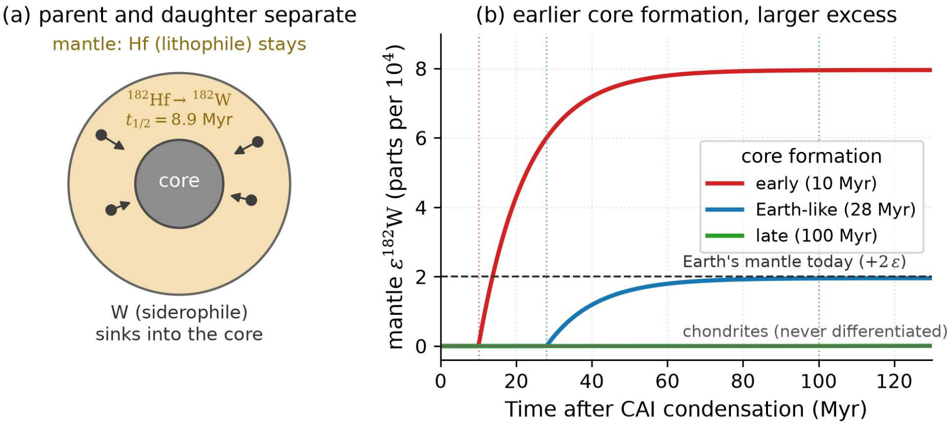
Both curves reach the band only near 40 to 60 GPa: the core equilibrated at the base of a 1000 to 1500 km deep magma ocean (Fischer et al. 2015).

Hf–W chronometry: when did cores form?

$^{182}\mathrm{Hf} \rightarrow {}^{182}\mathrm{W}$, half-life 8.9 Myr. Hf is lithophile, W siderophile: a core formed while $^{182}\mathrm{Hf}$ was alive leaves the mantle with excess $^{182}\mathrm{W}$; a late core leaves it chondritic, $\varepsilon^{182}\mathrm{W} \approx 0$.

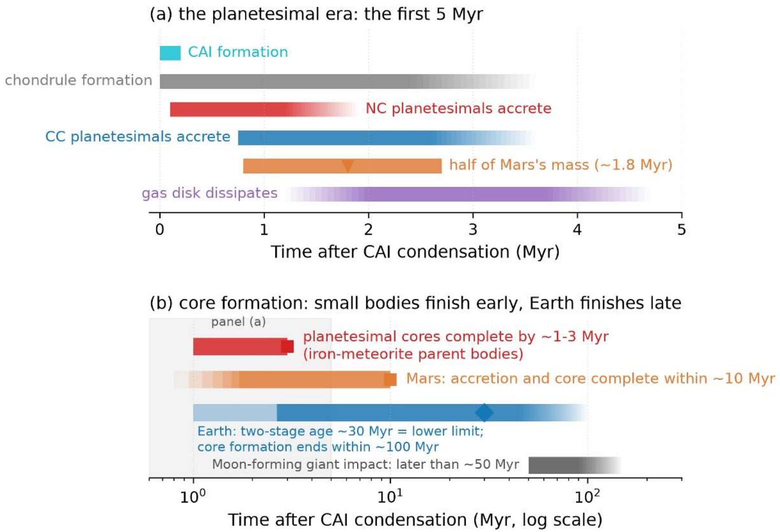
Body	ε^{182}W	Core formation timing
Earth	+2	two-stage ~30 Myr (lower limit), done by ~100 Myr
Mars	+0.4 bulk, +1.8 shergottites	complete within ~10 Myr
Moon	~Earth	consistent with the giant impact

Source: Kleine et al. (2009); Nimmo & Kleine (2015); Kleine & Walker (2017)

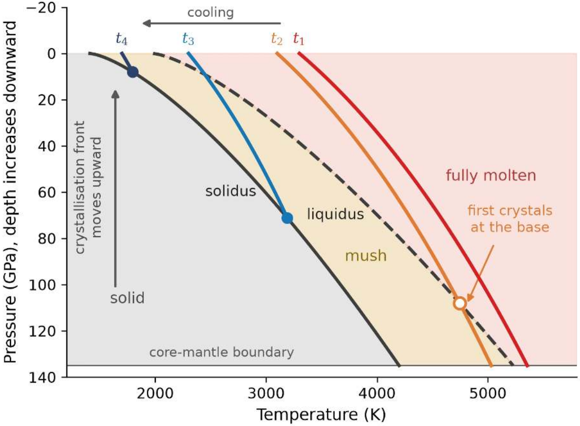


(a) Core formation separates Hf (mantle) from W (core); (b) mantle ^{182}W against time after CAI condensation for several core-formation ages. Course figure.

Earth's mantle at +2 ε gives a two-stage age of 28 to 30 Myr, a lower limit; protracted accretion places completion within about 100 Myr.

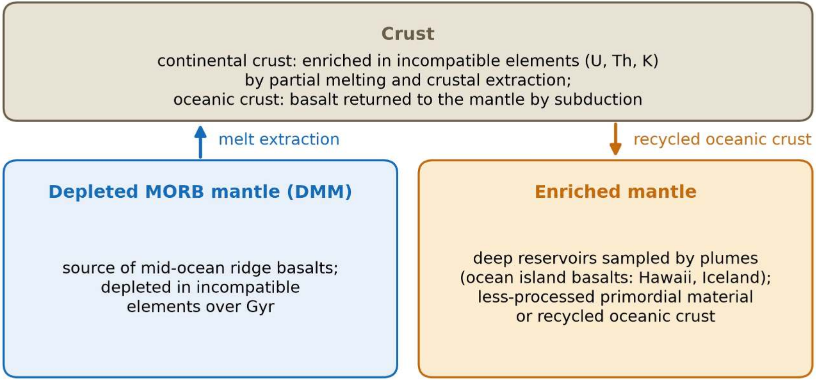


Time line from CAI condensation: planetesimal cores by 1 to 3 Myr, Mars within 10 Myr, Earth's core within 100 Myr. Course figure, after Lichtenberg et al. (2023).



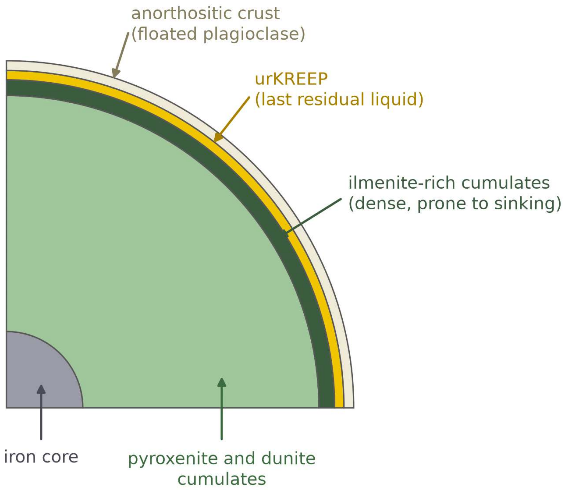
Peridotite solidus and liquidus against magma-ocean adiabats at successive times t_1 to t_4.
Course figure, after Elkins-Tanton (2012).

Melting curves steepen faster with pressure than adiabats, so deep magma oceans crystallise from the bottom up; dense melt can pool at the base.



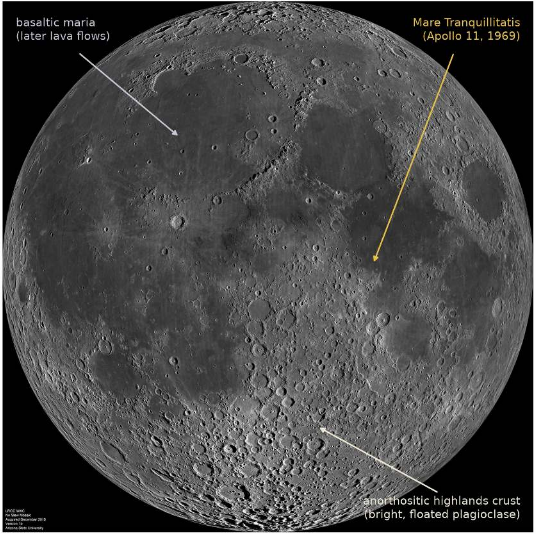
Mantle reservoirs after magma ocean crystallisation and later melting: depleted MORB mantle under the ridges, enriched mantle sampled by plumes. Course schematic.

U, Th and K concentrate in the melt and the crust, half of Earth's radiogenic heat; ocean island basalts record long-lived mantle heterogeneity.

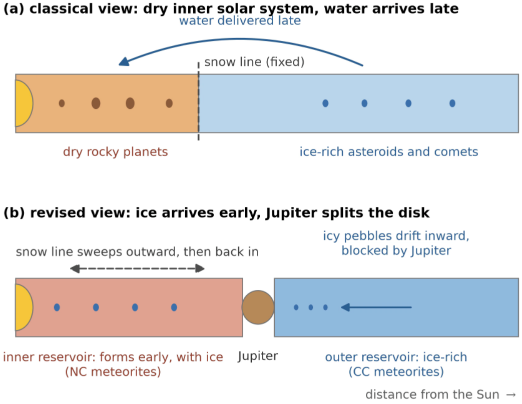


End-state of the lunar magma ocean: anorthosite flotation crust, the urKREEP layer, cumulates, and the dense ilmenite layer that later sinks. Course figure, after Elkins-Tanton (2012).

Anorthosite floated to the top about 4.4 Ga (Apollo samples 4.36 to 4.44 Ga) and **KREEP** is the last 1% of liquid: the Moon is the textbook case of magma ocean differentiation.



The bright highlands are the anorthositic flotation crust; the dark maria are later lava flows.
Credit: NASA/GSFC/Arizona State University (LROC WAC mosaic), public domain.



Classical (a) against revised (b) picture of volatile delivery. Course figure, after Alexander et al. (2019), Lichtenberg et al. (2021), Grewal et al. (2024).

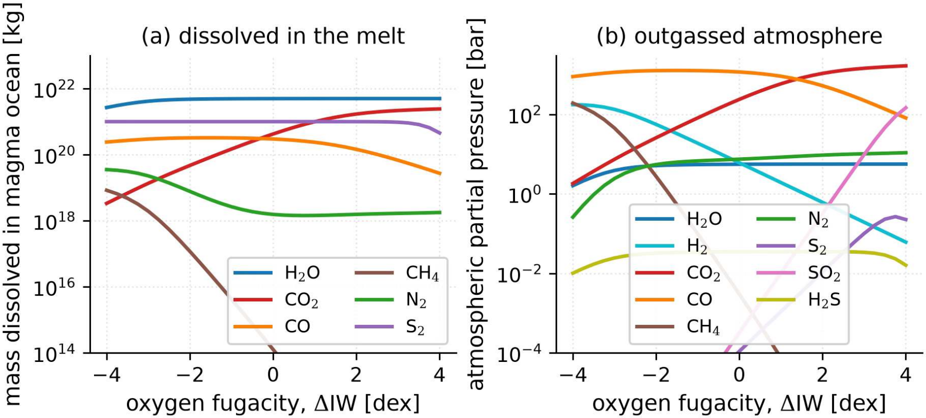
The snow line sweeps out and back, so the earliest inner planetesimals accrete ice, and Jupiter splits the disk into NC and CC: Earth's water was present at accretion.

Outgassing and volatile solubility

- Speciation follows the **oxygen fugacity** f_{O_2}: reducing gives H_2, CO, N_2, CH_4; oxidising gives H_2O, CO_2, N_2, SO_2
- Water dissolves as two OH groups per molecule, so $X_{H_2O} \propto p_{H_2O}^{1/2}$; CO_2 is about 100× less soluble
- A thicker steam atmosphere lets the magma keep more water: a self-limiting feedback

A planet cannot outgas its entire water budget; the mantle redox state at solidification sets the initial atmosphere (Lectures 5–6).

Source: Hirschmann (2012)



Volatile partitioning between magma ocean and atmosphere against oxygen fugacity (Nicholls et al. 2024; bulk silicate Earth inventory of Krijt et al. 2023).

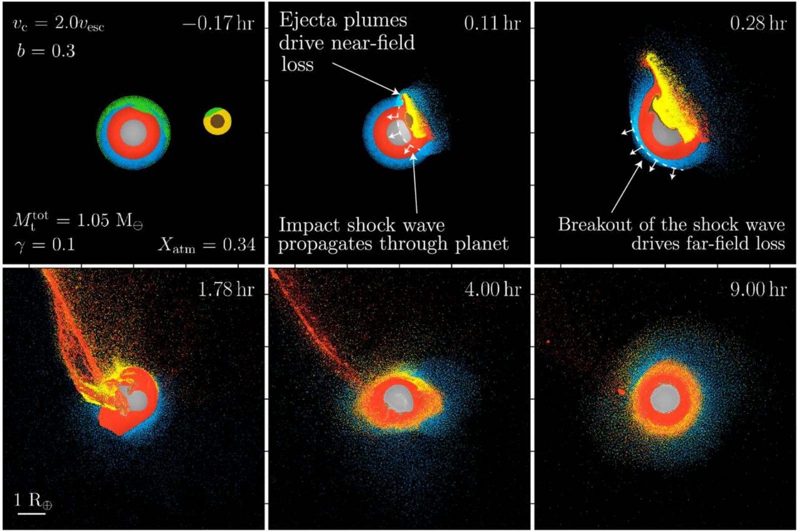
The melt keeps almost all the water dissolved (over 3 oceans) while carbon fills the atmosphere at hundreds of bar: most of the water sits in the magma.

Impact erosion and delivery

- **Local:** an ordinary impactor removes at most the air above the tangent plane, $H/2R \approx 6 \times 10^{-4}$ of Earth's atmosphere per event, cumulative
- **Global:** even the Moon-forming impact removes only about 20%, more with a surface ocean
- The same planetesimals deliver volatiles; Venus keeps about 50× more $^{36}\mathrm{Ar}$ than Earth

Earth's air and oceans are the ledger of delivery minus erosion, closed by magma-ocean outgassing.

Source: Melosh & Vickery (1989); Genda & Abe (2003, 2005); Schlichting et al. (2015)



Atmospheric loss in a giant impact: ejecta plumes drive near-field loss, the far-side shock breakout drives far-field loss; green particles are lost. Roche et al. (2025), CC BY 4.0.

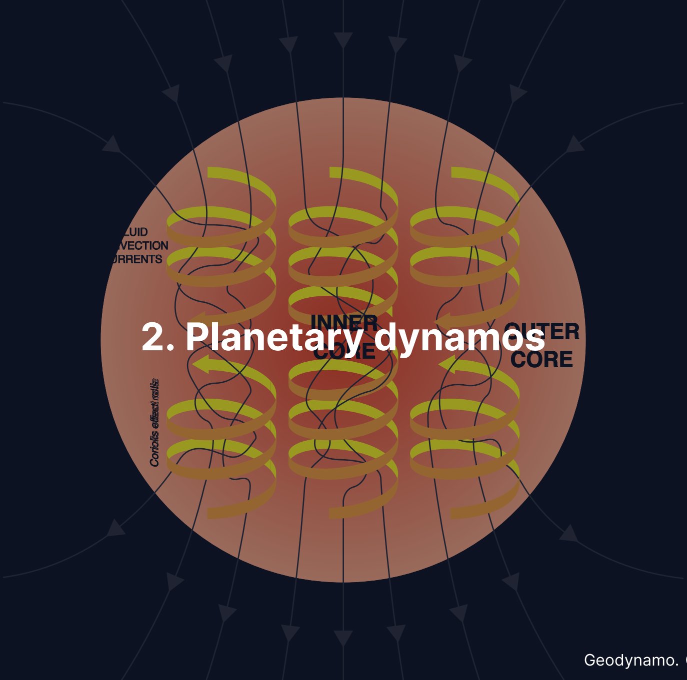
Meteoritic dichotomy: NC vs. CC

- **Non-carbonaceous (NC)** and **carbonaceous (CC)** meteorites have distinct $^{50}\mathrm{Ti}$, $^{54}\mathrm{Cr}$ and $^{48}\mathrm{Ca}$ ratios
- Two reservoirs in the early nebula: **Jupiter** formed early and blocked mixing between inner and outer disk
- Both reservoirs fed Earth's accretion, in changing proportions over time

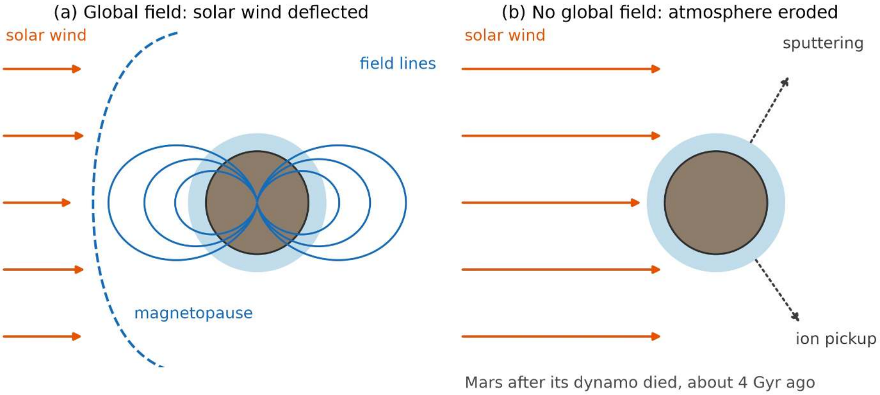
The dichotomy is one of the strongest meteoritic constraints on the timing of giant planet formation.

Source: Kruijer et al. (2017); Lichtenberg et al. (2021)

2. Planetary dynamos



Geodynamo. Credit: Wikimedia, CC BY-SA 4.0



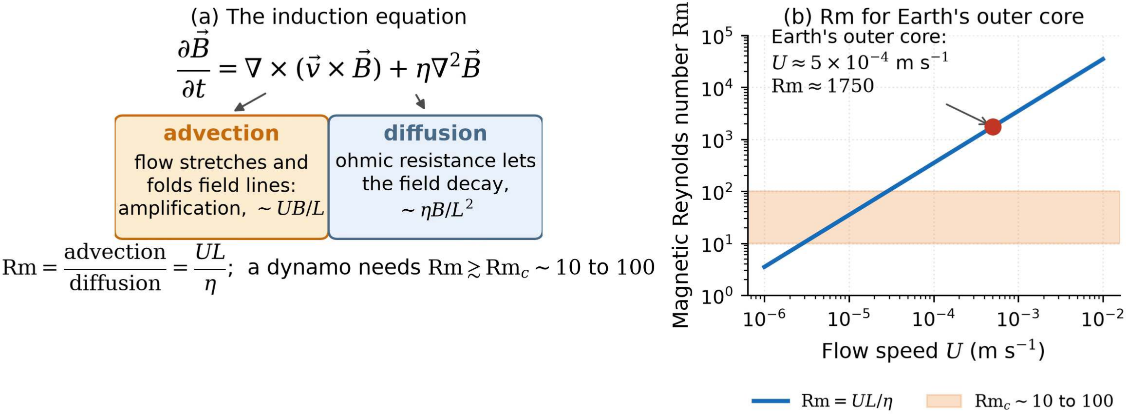
A global field as a shield: (a) a magnetised planet holds the solar wind off at the magnetopause; (b) without one, sputtering and ion pickup erode the upper atmosphere. Course schematic.

A field deflects the solar wind and limits the cosmic-ray dose; Mars lost its field about 4 Ga and then about 0.8 bar of atmosphere (MAVEN).

The induction equation and the three requirements for a dynamo

$$\frac{\partial \vec{B}}{\partial t} = \underbrace{\nabla \times (\vec{v} \times \vec{B})}_{\text{advection: flow amplifies the field}} + \underbrace{\eta \nabla^2 \vec{B}}_{\text{diffusion: ohmic decay}}, \qquad \eta = \frac{1}{\mu_0 \sigma}$$

1. **Electrically conducting fluid:** liquid iron alloy, or metallic hydrogen in the gas giants
2. **Convection:** thermal (core hotter than mantle) or compositional (inner-core crystallisation)
3. **Sufficient flow vigour:** advection beats diffusion, $\mathrm{Rm} \gtrsim 10$–100

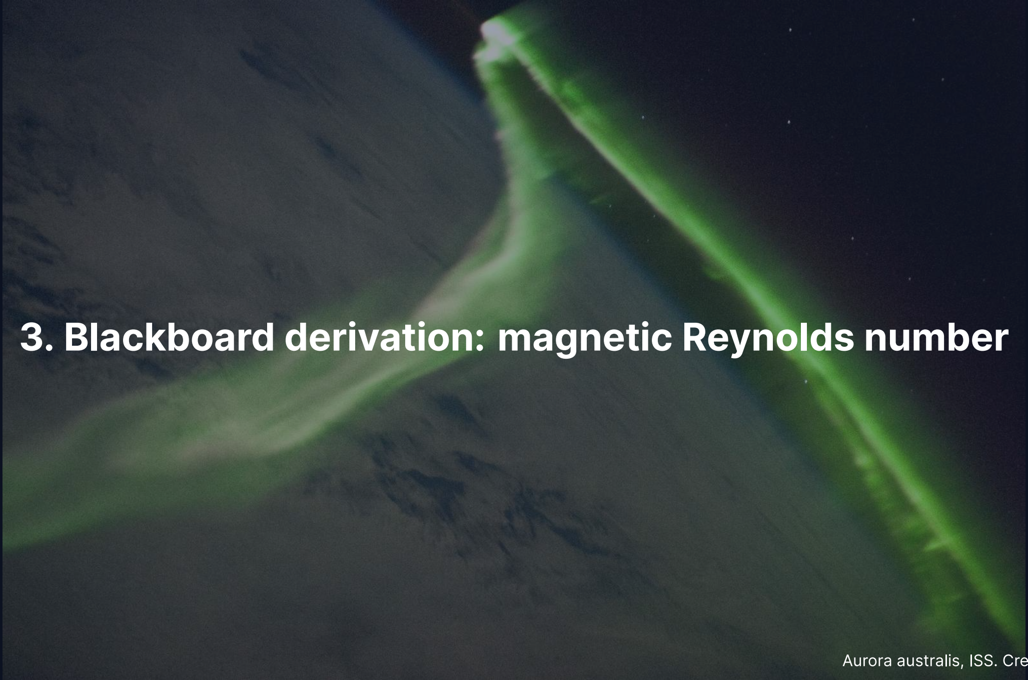


(a) Advection at ~ UB/L against diffusion at ~ $\eta B/L^2$; (b) Rm for Earth's outer core against flow speed. Course figure.

Rm = UL/η must exceed a critical value of order 10 to 100; at $U \approx 5 \times 10^{-4}$ m s^{-1} Earth's core reaches about 1750.

Break

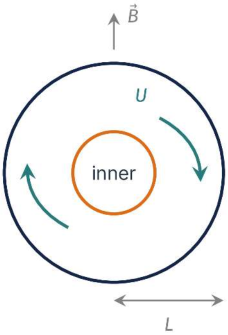
Earthrise, Apollo 8 (1968). Credit: NASA



3. Blackboard derivation: magnetic Reynolds number

Aurora australis, ISS. Credit: NASA

Goal and strategy



Goal: does advection or diffusion win in Earth's core?

- Estimate each term of the induction equation by scales
- Take their ratio: the field strength cancels
- Put in the numbers for the outer core

A conducting shell of size L, flow speed U, diffusivity η: one dimensionless number decides.

Deriving the magnetic Reynolds number

Dimensional analysis of each term: advection $|\nabla \times (\vec{v} \times \vec{B})| \sim UB/L$, diffusion $|\eta \nabla^2 \vec{B}| \sim \eta B/L^2$.

$$\mathrm{Rm} = \frac{\text{advection}}{\text{diffusion}} = \frac{UB/L}{\eta B/L^2}, \qquad B \text{ cancels:}$$

$$\boxed{\mathrm{Rm} = \frac{UL}{\eta}}$$

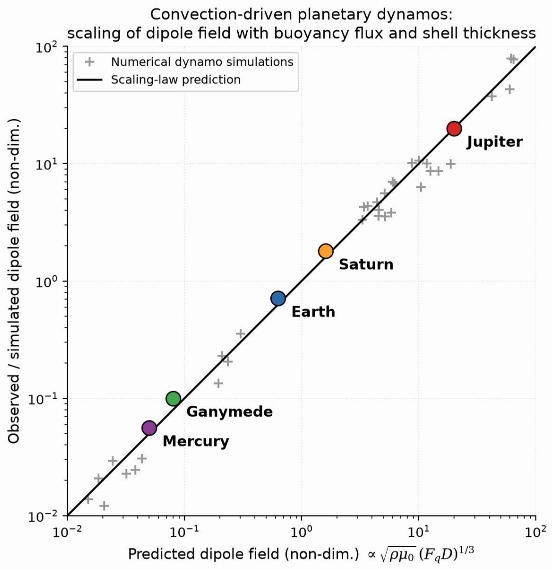
$\mathrm{Rm} \gg 1$: the field is "frozen in" to the flow; $\mathrm{Rm} \ll 1$: the field decays; critical $\mathrm{Rm}_c \sim 10$–100. Rm depends on the **flow**, not on the field.

Application: Earth's outer core

Parameter	Symbol	Value
Core radius	L	3.5×10^6 m
Flow velocity	U	5×10^{-4} m s^{-1}
Magnetic diffusivity	η	~1 m^2 s^{-1}

$$\mathrm{Rm}_\oplus = \frac{5 \times 10^{-4} \times 3.5 \times 10^6}{1} \approx 1750 \gg \mathrm{Rm}_c$$

Ohmic diffusion time $\tau_{\mathrm{ohm}} \sim L^2/\eta \approx 400{,}000$ yr $\ll$ Earth's age: the field must be **continuously regenerated**.



Dynamo simulations collapse onto one power law that predicts Earth, Jupiter, Mercury and Ganymede. Course figure, after Christensen & Aubert (2006).

$B_{\mathrm{dip}} \propto \sqrt{\rho\mu_0}\,(F_q D)^{1/3}$ (F_q buoyancy flux, D shell thickness): field strength is set by the **available convective power**, not by rotation rate; Venus is absent, its dynamo off.

Hf–W chronometry: beyond the first estimate

- $D_W^{\mathrm{met/sil}}$ falls from about 40 at the surface to about 10 at 60 GPa: the permitted window widens to 30–100 Myr
- Pd–Ag and Pt–Os cross-check the timing
- **Disequilibrium core formation:** not all metal re-equilibrates before sinking; Earth's core formed in episodes

Source: Rudge et al. (2010); Kleine et al. (2017)

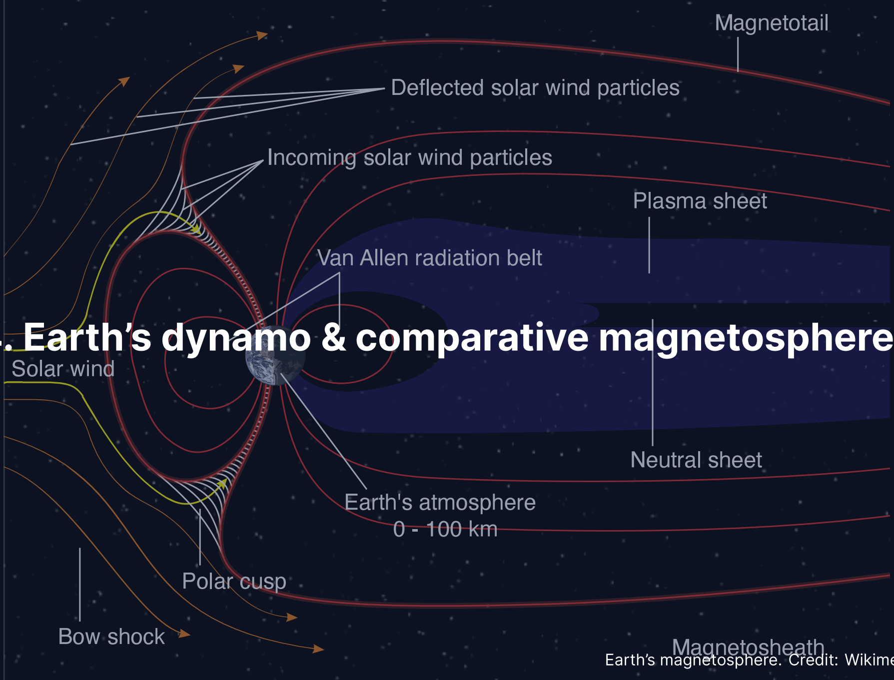
How old is Earth's magnetic field?

- Oldest confirmed field 3.4–3.5 Gyr (Archean South Africa); single zircons push it to 4.2 Gyr (Tarduno et al. 2015)
- Continuous from ≳3.4 Ga to today: komatiites, banded iron formations
- Proterozoic paleointensities near modern; polar wander paths trace supercontinent cycles

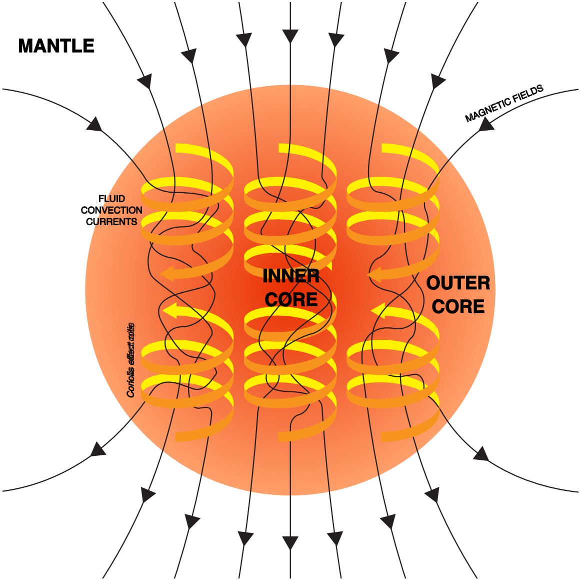
Earth's dynamo has run for at least 3.5 Gyr, possibly since the Hadean, shielding the early atmosphere while life was emerging.

Source: Tarduno et al. (2010, 2015); Biggin et al. (2015)

4. Earth’s dynamo & comparative magnetospheres



Earth’s magnetosphere. Credit: Wikimedia, CC BY-SA 3.0

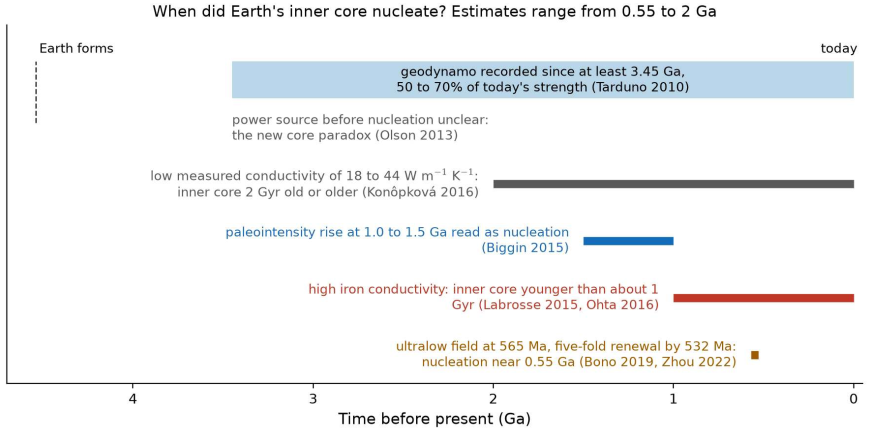


The geodynamo: convection columns in the liquid outer core organised by rotation. Credit: Wikimedia, CC BY-SA 4.0.

The liquid Fe alloy of the outer core (1220 to 3480 km from the centre) convects thermally and compositionally: a mainly **dipolar** field tilted about 11° from the spin axis, 25 to 65 μT at the surface.

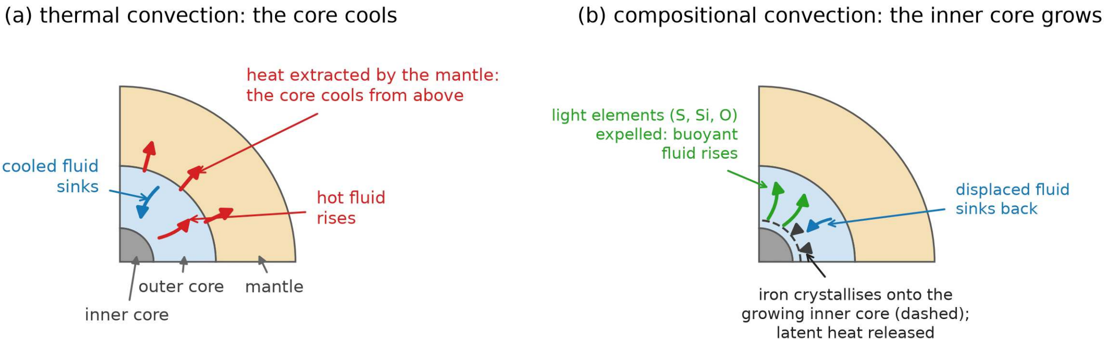
Earth's core: composition & growth

- **Inner core** (0–1220 km): solid Fe–Ni, growing at about 0.5 mm yr^{-1}; its age is debated, 0.55 to 2 Gyr
- **Outer core** (1220–3480 km): liquid Fe–Ni, 5–10% less dense than pure Fe from light elements (S, Si, O, C, H)
- Inner-core crystallisation ejects light elements: **compositional convection** drives today's dynamo

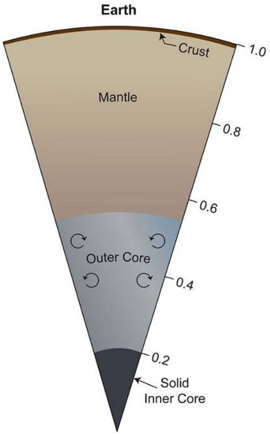


The inner-core age debate on a time line: iron conductivity, the paleointensity record, and the new core paradox. Course figure.

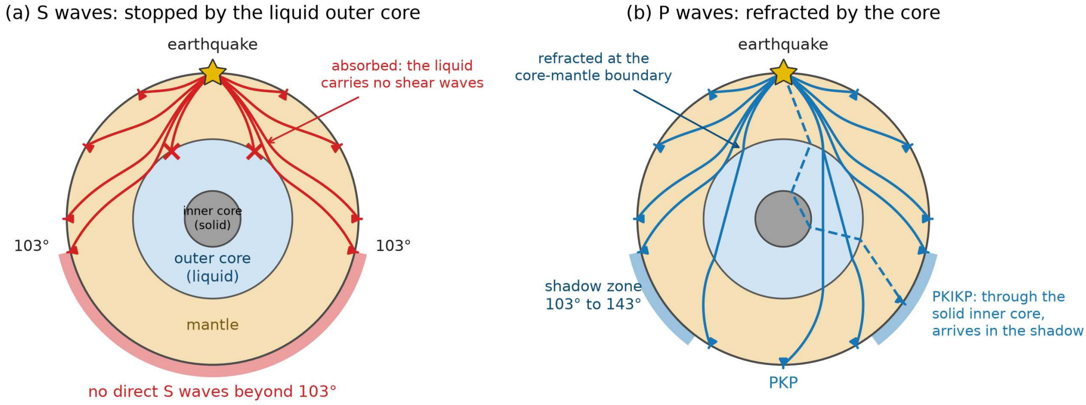
Nucleation estimates run from 0.55 to 2 Ga; a field existed at 3.45 Ga, and what powered it before the inner core is the **new core paradox**.



The two buoyancy sources of the geodynamo: (a) cooling from above across the core-mantle boundary, (b) light elements and latent heat released by inner-core growth. Course figure.

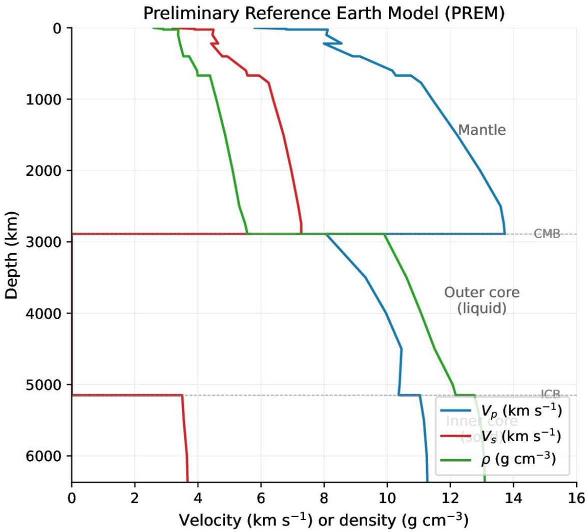


Cross-section of Earth's interior. Credit: NASA/JPL-Caltech/SwRI (PIA25063).



Seismic shadow zones: *S* waves stop at the outer core, *P* waves refract into a shadow from 103° to 143°. Course figure.

No direct *S* beyond 103°: the outer core is **liquid** (Oldham 1906); weak PKIKP arrivals inside the *P* shadow reveal a **solid inner core** (Lehmann 1936).

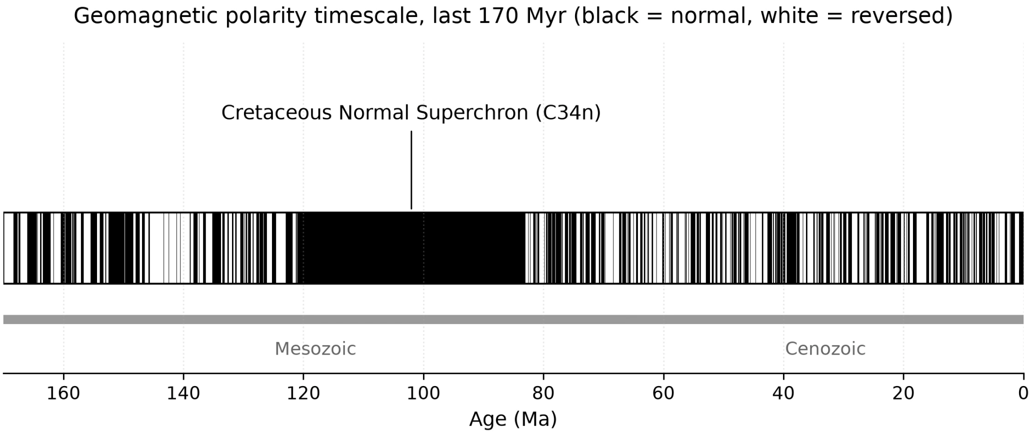


PREM velocities and density against depth: V_s vanishes between 2890 and 5150 km, and the density jumps at both core boundaries. Course figure, after Dziewonski & Anderson (1981).

Secular variation of the field

- **Secular variation** (decades to centuries): non-dipole components grow, decay and drift
- **Westward drift** at about 0.2° yr^{-1}: differential rotation between core and mantle
- **South Atlantic Anomaly:** about 30% weaker field, the surface expression of a reversed-flux patch at the core–mantle boundary

Source: Finlay et al. (2020), Swarm satellite constellation data



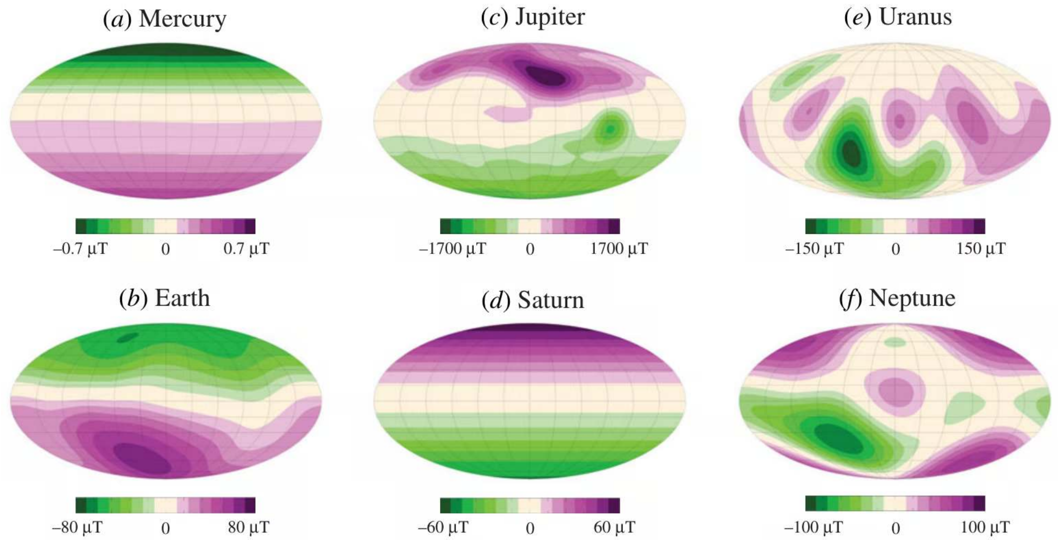
The geomagnetic polarity time scale. Course figure; chron ages after Gradstein et al. (2020).

Poles **swap** 4 to 5 times per Myr, each over 1000 to 10 000 yr; the Cretaceous Normal Superchron held one polarity for 38 Myr; **magnetic stripes** on the seafloor record spreading.

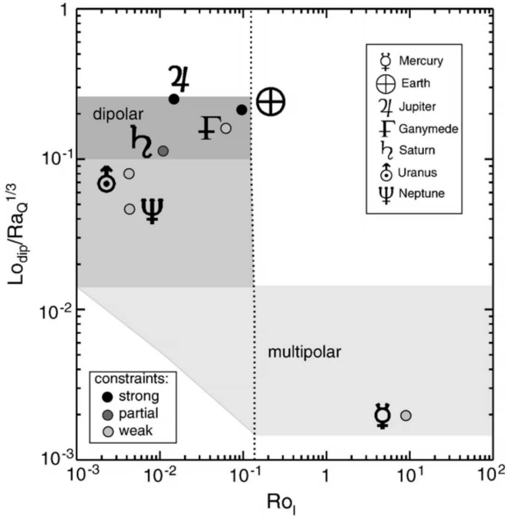
Magnetic fields across the solar system

Body	Field type	Surface field	Rel. dipole
Earth	Active dynamo	25–65 μT	1
Mercury	Active (weak)	~0.3 μT	5×10^{-4}
Venus	None detected	<0.01 μT	$< 10^{-5}$
Mars	Remnant crustal	up to 1500 nT	n/a
Jupiter	Active (strong)	400–1400 μT	20,000
Saturn	Active	~20 μT	~600
Ganymede	Active	~0.7 μT	1.5×10^{-3}

Jupiter's dipole moment is about 20,000× Earth's, generated in metallic hydrogen.



Surface radial fields of six dynamos: axial dipoles at Earth, Jupiter and Saturn; tilted multipoles at Uranus and Neptune; a weak, offset dipole at Mercury. Soderlund & Stanley (2020).



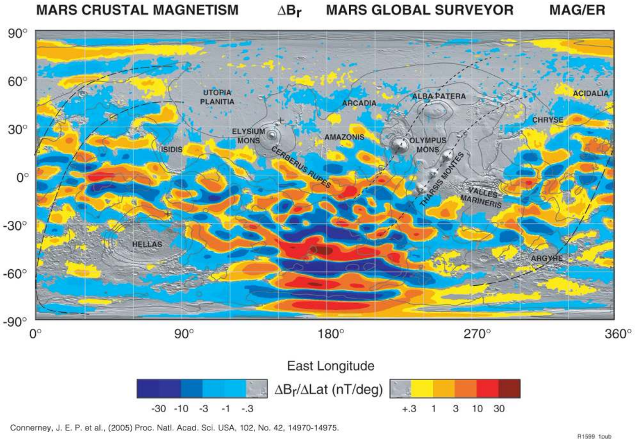
Dipolar against multipolar dynamos in the local Rossby number: Earth, Jupiter, Saturn and Ganymede dipolar; Mercury, Uranus and Neptune near the multipolar regime. Olson & Christensen (2006).

Venus and Mercury: two puzzles

- **Venus, no field:** slow rotation alone is insufficient; no inner core, and a stagnant lid that extracts core heat too slowly
- **Mercury, an offset dynamo:** 0.3 μT, 1% of Earth's, offset 500 km north (MESSENGER)
- A thin liquid shell above a partly solid deeper core limits Mercury's convective vigour

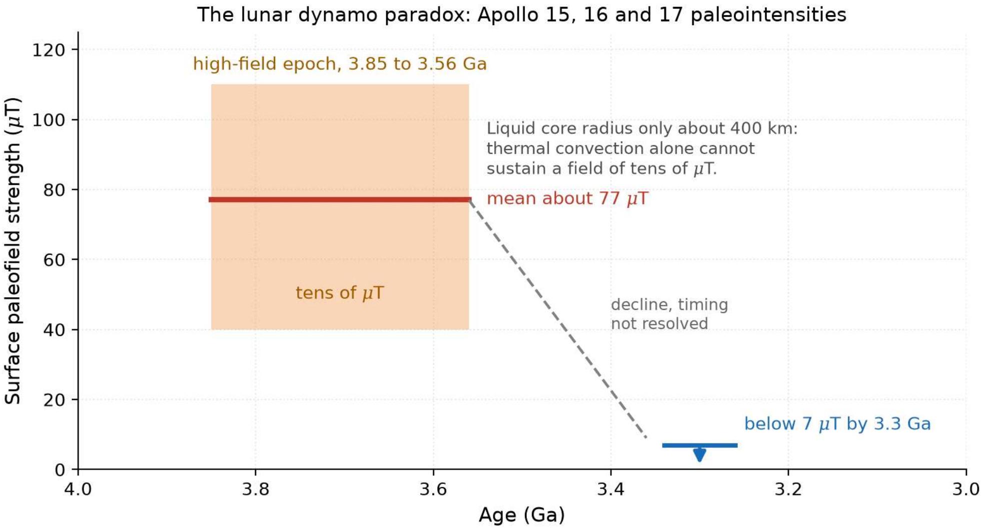
Venus most likely combines reduced core cooling (no plate tectonics) with no inner-core nucleation.

Source: Anderson et al. (2012) for Mercury

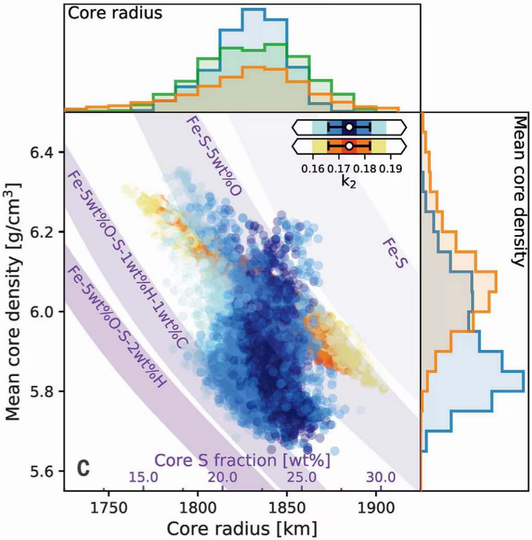


Crustal magnetisation of Mars: intense in the southern highlands, absent in the northern lowlands and the large basins. Credit: NASA/GSFC.

Mars **had** a dynamo that died between about 4.1 and 3.7 Ga (Mittelholz et al. 2020); MAVEN measures 2 to 3 kg s^{-1} of loss today, about 0.8 bar over 4 Gyr.



The lunar dynamo paradox: Apollo paleofields of about 77 μT from 3.85 to 3.56 Ga, below 7 μT by 3.3 Ga, from a core of only about 400 km (Weiss & Tikoo 2014). Course figure.



InSight's Mars core: $R_{\mathrm{core}} \approx 1830$ km and a density near 6 g cm^{-3}, too light for pure iron; a sulfur-rich core may never nucleate an inner core. Stähler et al. (2021).

Jupiter and Ganymede: two more dynamos

- **Jupiter:** H_2 becomes **metallic** above about 100 GPa, at 0.85 Jupiter radii; the strongest field in the solar system
- **Ganymede:** the only moon with an intrinsic field, 0.7 μT (Galileo 1996), sustained by tidal heating or a slowly freezing core
- JUICE (arriving 2031) will map Ganymede's field in detail

Source: Connerney et al. (2022); Bolton et al. (2017); Kivelson et al. (1996); Bland et al. (2008)

Body	Dynamo today	What the field tells us
Mercury	active	weak, about 1% of Earth's; dipole offset northward; thin liquid shell
Venus	none detected	no inner core to drive compositional convection; stagnant lid slows core cooling
Earth	active	reference case; inner-core nucleation within the past 1 Gyr
Moon	extinct	paleofields of tens of microtesla from 3.85 to 3.56 Ga, below 7 microtesla by 3.3 Ga
Mars	extinct	remnant crustal magnetism in the southern highlands; global field lost about 4 Gyr ago
Ganymede	active	the only moon with an active dynamo; surface field about 0.7 microtesla, above Mercury's 0.3 microtesla

Dynamo status of the terrestrial bodies and Ganymede: Mercury weak and offset, Venus none, Earth active, the Moon and Mars extinct, Ganymede active. Course figure.

Magnetopause standoff distance

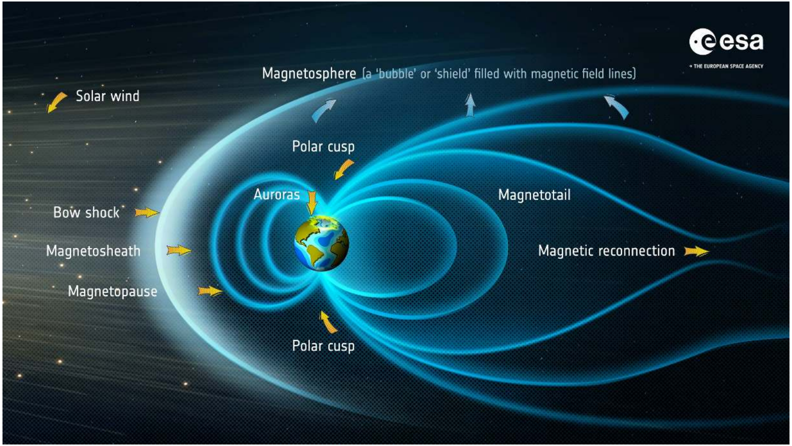
Pressure balance at the magnetopause:

$$\underbrace{\frac{1}{2}\rho_{\mathrm{sw}} v_{\mathrm{sw}}^2}_{\text{solar wind ram pressure}} = \underbrace{\frac{B_{\mathrm{mp}}^2}{2\mu_0}}_{\text{magnetic pressure}}$$

For a dipole, $B \propto r^{-3}$, so the standoff distance scales as

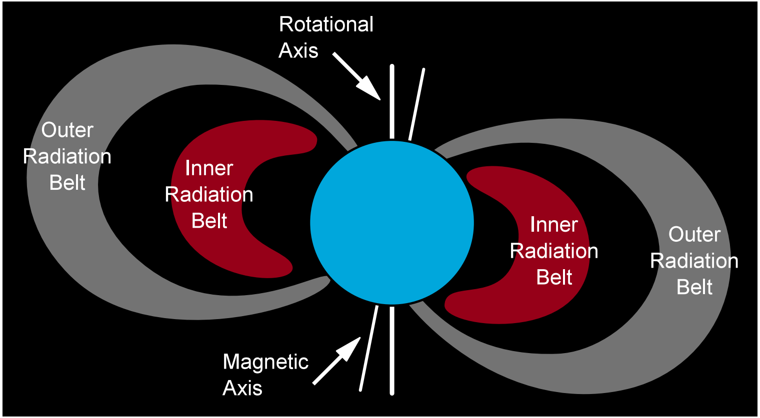
$$r_{\mathrm{mp}} \propto \left(\frac{B_0^2}{\mu_0 \rho_{\mathrm{sw}} v_{\mathrm{sw}}^2}\right)^{1/6}$$

- Earth under typical solar wind: $r_{\mathrm{mp}} \approx 10\, R_{\oplus}$; compressed to about 6 $R_{\oplus}$ in a CME
- Jupiter: 50–100 R_J



Anatomy of Earth's magnetosphere: bow shock, magnetosheath, the magnetopause at about $10\,R_\oplus$, and the stretched nightside tail. Credit: ESA, CC BY-SA 3.0 IGO.

In the tail, opposing field lines **reconnect** and release stored magnetic energy: earthward particles make **aurorae** and substorms, downstream **plasmoids** leave.



Cross-section of Earth’s Van Allen radiation belts. Credit: Booyabazooka / NASA, public domain.

Earth’s dipole traps energetic particles in two belts (Explorer 1, 1958): 10–100 MeV protons at 1.5 $R_{\oplus}$ and 0.1–10 MeV electrons at 4–5 $R_{\oplus}$, a hazard for spacecraft and crews.

Aurora australis seen from the International Space Station. Credit: NASA/ISS.

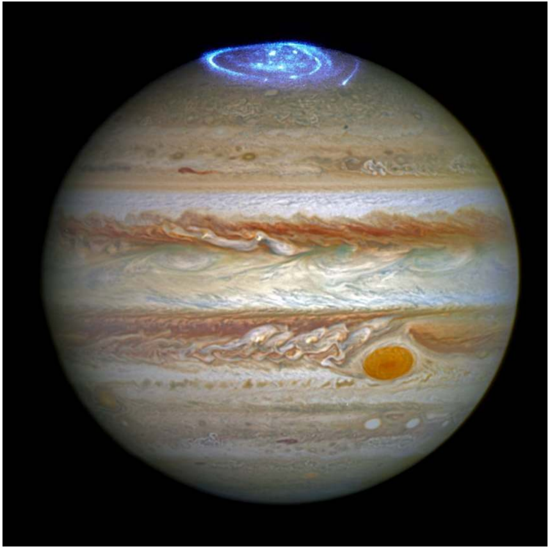
Particles precipitate along field lines into the polar atmosphere: **green** O at 100–200 km (557.7 nm), **red** O at 200–400 km (630.0 nm), **blue** N_2 bands; every magnetised planet with an atmosphere has aurorae.

Why aurorae happen at the poles, and Jupiter's

- The dipole focuses particles toward the magnetic poles; the polar **cusps** give the solar wind direct access
- Electrons accelerated by tail reconnection reach 100–400 km at 1–100 keV; the lines are fingerprints of the excited species
- **Jupiter:** 10^{13}–10^{14} W of UV, powered by its 10 h rotation and about 1 tonne s^{-1} of sulfur and oxygen from Io; the moons leave **auroral footprints**

Aurorae map the power a magnetosphere dumps into its atmosphere: solar-wind coupling at Earth, rotation and Io at Jupiter.

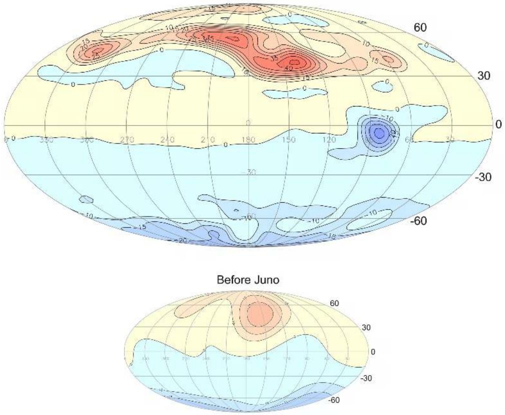
Source: Connerney et al. (2022); Mauk et al. (2017)



Jupiter’s ultraviolet aurora over an optical image, Hubble Space Telescope. Credit: NASA, ESA, J. Nichols (University of Leicester).

5. Recent advances & outlook

Aurora australis. Credit: NASA/ISS



Jupiter's surface radial field after Juno (top) and before (bottom): outward flux in yellow and red, inward in blue. Credit: NASA/JPL-Caltech/SwRI (PIA25040).

Juno finds a strongly **non-dipolar** field with a northern flux lobe and the reversed Great Blue Spot near the equator: the dynamo runs in a shallow shell above the dilute core (Lecture 8).

Missions to cores and dynamos

- **BepiColombo** (ESA/JAXA): flybys 2021–2025 confirmed Mercury's weak, offset dipole; polar-orbit mapping after the late-2026 insertion
- **JUICE** (Jupiter 2031, Ganymede orbit 2034): separates Ganymede's intrinsic field from the induced ocean signature
- **Psyche** (arrival 2029): a 220 km M-type asteroid, a candidate exposed planetesimal core, tested for remnant magnetisation

Source: Benkhoff et al. (2021); Grasset et al. (2013); Elkins-Tanton et al. (2017)

Open questions in planetary magnetism

- **When did the inner core nucleate**, and how did Earth sustain a dynamo before it?
- **Why does Venus have no field**, and why is Mercury's so weak yet active?
- **How important is a magnetic field for habitability?**

Looking ahead to Lecture 5

A planet with a core, a mantle, a magnetic field, and a freshly outgassed atmosphere.

- Lecture 5 (dr. Mara Attia): primary, secondary and tertiary atmospheres
- Hydrostatic equilibrium, the greenhouse effect and the energy balance
- Atmospheric escape: magnetic shielding is one control, thermal escape proceeds regardless

Differentiation builds the planet; the atmosphere it outgasses is where climate and habitability are decided.

Summary

1. Magma oceans drive **metal–silicate separation**; Hf–W gives a two-stage age of about 30 Myr for Earth's core, a lower limit, with completion within about 100 Myr
2. A dynamo needs a conducting fluid, convection and $\mathrm{Rm} \gtrsim 10$–100; Earth's $\mathrm{Rm} \approx 1750$
3. Mars lost its dynamo between about 4.1 and 3.7 Ga and then its atmosphere; Venus has no detectable field

Questions?

Next: Lecture 5, Atmospheres I: composition, structure & energy balance (Mara Attia)

Mon 21 Sep, 15:00–17:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

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